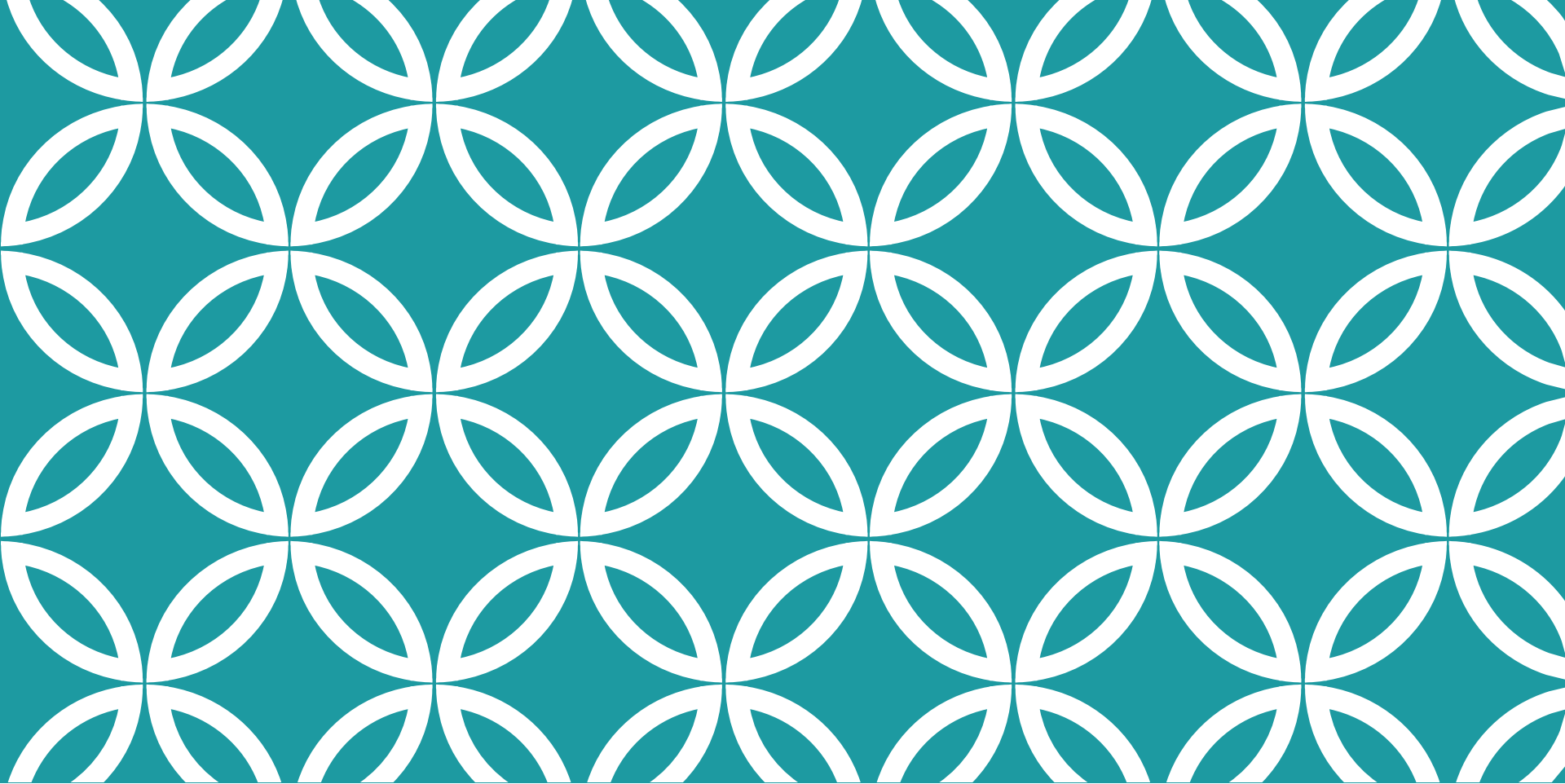


SPEECH SIGNAL REPRESENTATION

Reference Text Books:

1. Fundamentals of Speech recognition – L. Rabiner and B. Juang, Prentice Hall signal processing series.
2. Discrete-time speech signal processing: principles and practice, Thomas F. Quatieri, Coth.
3. Digital Signal Processing John G. Proakis Dimitris K. Manolakis



LINEAR TIME-INVARIANT (LTI) SYSTEMS



DISCRETE TIME SYSTEMS

A *discrete-time system* is a system that transforms an input sequence ($x[n]$) to an output sequence $y[n]$,

$$y[n] = T(x[n]).$$

Example: Microphone $\rightarrow$ converts acoustic speech into electrical signal.

In **speech processing**, discrete-time systems model the vocal tract filter.

Linear time-invariant (LTI) systems

When we restrict this transformation to have the properties of **linearity and time invariance**, we form a class of *linear time-invariant* (LTI) systems.

LINEAR TIME INVARIANT (LTI) SYSTEMS

Time-invariance

A system is *time-invariant* if a time shift in the input by *no* samples gives a shift in the output by *no* samples, i.e.,

$$\begin{aligned} \text{if } y[n] &= T(x[n]), \\ \text{then } y[n - n_0] &= T(x[n - n_0]) \end{aligned}$$

Example: Let's consider the following transformation

$$y[n] = x[n] + x[n - 1]$$

$$\text{Input: } x[n] = \{1, 2, 3\}$$

$$\text{System output: } y[n] = \{0 + 1, 1 + 2, 2 + 3\} = \{1, 3, 5\}$$

$$\text{Shift input by } n_0 = 1$$

R. H. S.

Output of shifted input:

$$T(x[n - 1]) = T(\{0, 1, 2\}) = \{0 + 0, 0 + 1, 1 + 2\} = \{0, 1, 3\}$$

L. H. S.

$$\text{Shifted system output: } y[n - 1] = \{0, 1, 3\}$$

The **role of time invariance** is to guarantee that shifting the input produces the same shift in the output, without altering the system's behaviour.

LINEAR TIME INVARIANT (LTI) SYSTEMS

Linearity - Principle of Superposition

Let $x_1[n]$ and $x_2[n]$ be the inputs to a discrete-time system.

For arbitrary constants a and b , the system is linear if and only if

$$y[n] = T(ax_1[n] + bx_2[n]) = aT(x_1[n]) + bT(x_2[n]) = ay_1[n] + by_2[n]$$

which is sometimes referred to as the “**principle of superposition**”.

Example: Let's consider the following transformation

$$y[n] = T(x[n]) = 2 * x[n]$$

Input 1: $x_1[n] = \{1, 2\} \rightarrow$ Output: $y_1[n] = 2 * x_1[n] = \{2, 4\}$

Input 2: $x_2[n] = \{3, 4\} \rightarrow$ Output: $y_2[n] = 2 * x_2[n] = \{6, 8\}$

Let's take $a = 3$, $b = 2$

L. H. S.

$$T(3x_1[n] + 2x_2[n]) = T(\{3*1 + 2*3, 3*2 + 2*4\}) = T(\{9, 14\}) = \{18, 28\}$$

R. H. S.

$$3 * y_1[n] + 2 * y_2[n] = \{3 * (2 * 1) + 2 * (2 * 3), 3 * (2 * 2) + 2 * (2 * 4)\} = \{18, 28\}$$

DISCRETE TIME SYSTEMS

Impulse Decomposition Property of Discrete-time Signal: Any discrete time signal, $x[n]$, can be represented by a weighted sum of delayed impulses, i.e.,

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n - k]$$

where $\delta[n - k]$ is a **shifted unit impulse** or **delayed delta function**.

The **unit impulse** $\delta[n]$ is defined as:

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

$n - k$ means the impulse has been **shifted** to occur at $n = k$. Each **impulse** $\delta[n - k]$ marks the position in time.

Example: $x[n] = \{x[0] = 3, x[1] = 4, x[2] = 5\}$

Then the decomposition can be written as

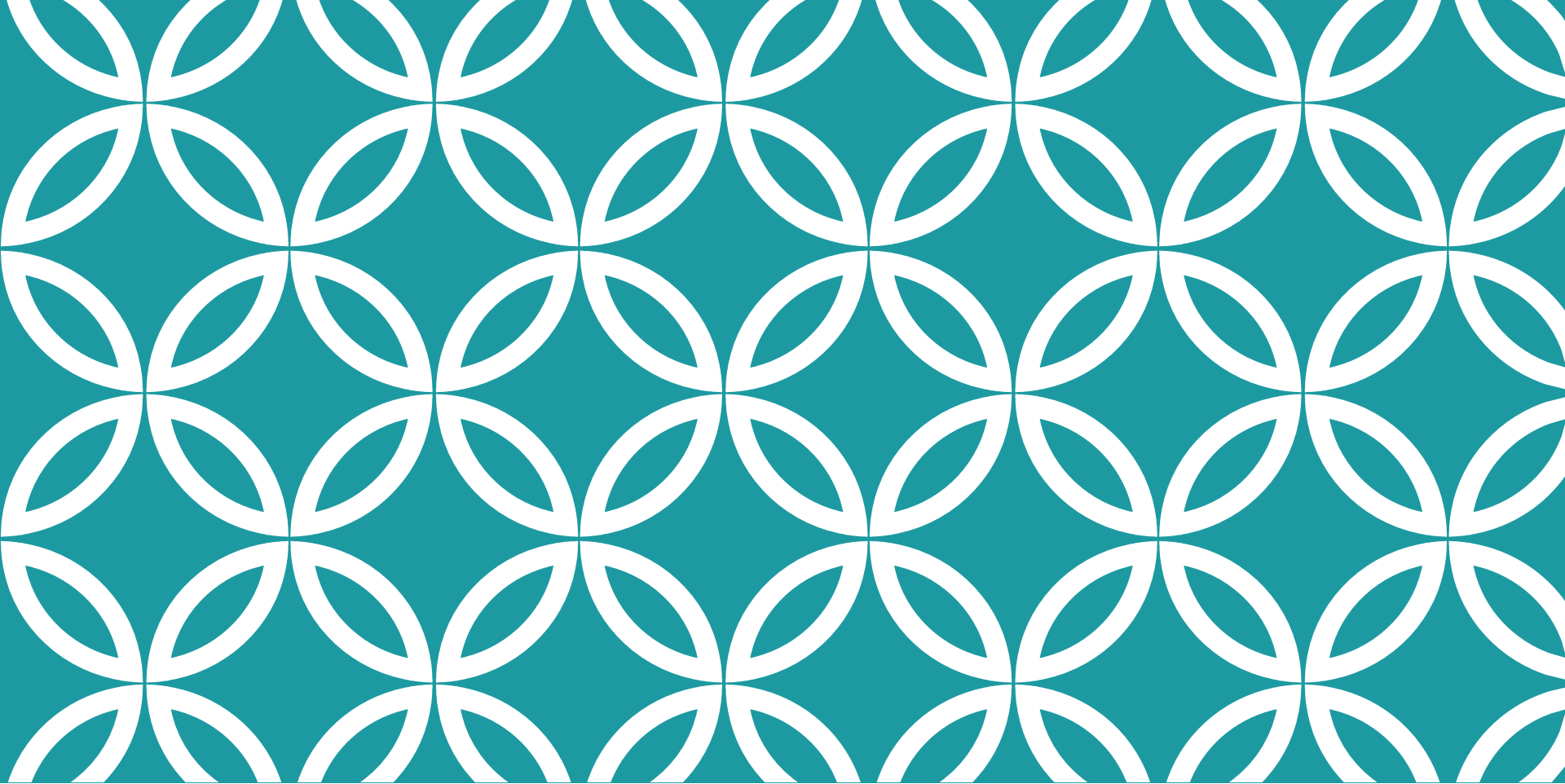
$$x[n] = 3\delta[n] + 4\delta[n - 1] + 5\delta[n - 2]$$

for $n = 0$, $3\delta[0] + 4\delta[-1] + 5\delta[-2] = 3(1) + 4(0) + 5(0) = 3$

for $n = 1$, $3\delta[1] + 4\delta[0] + 5\delta[-1] = 3(0) + 4(1) + 5(0) = 4$

for $n = 2$, $3\delta[2] + 4\delta[1] + 5\delta[0] = 3(0) + 4(0) + 5(1) = 5$

Signal is successfully reconstructed $\{3, 4, 5\}$



CONVOLUTION IN TIME DOMAIN |

IMPULSE RESPONSE

An important property of an LTI system is that it can be completely characterized by its impulse response.

Impulse response is defined as a system's response to a unit sample (or an impulse signal).

Given an LTI system, the output $y[n]$ for an input $x[n]$ is given by a sum of weighted and delayed impulse responses, i.e.,

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = x[n] * h[n]$$

where $h[n]$ is the impulse response of the system

Each input sample $x[k]$ excites the system's impulse response $h[n]$.

This is referred to as **the convolution of $x[n]$ with $h[n]$** , where “*” denotes the convolution operator.

IMPULSE RESPONSE

Example 1:

Input signal: $x[n] = \{1, 2, 1\}$ (at $n = 0, 1, 2$)

Impulse response: $h[n] = \{1, -1\}$ (at $n = 0, 1$)

$$y[0] = 0 * h[1] + x[0]h[0] = 1 \cdot 1 = 1$$

$$y[1] = x[0]h[1] + x[1]h[0] = 1(-1) + 2(1) = -1 + 2 = 1$$

$$y[2] = x[1]h[1] + x[2]h[0] = 2(-1) + 1(1) = -2 + 1 = -1$$

$$y[3] = x[2]h[1] + 0 * h[0] = 1(-1) = -1$$

So, the output $y[n] = \{1, 1, -1, -1\}$

IMPULSE RESPONSE

Example 2:

Input signal: $x[n] = \{2, 1, 2, 4, 3\}$ at $n = 0, 1, 2, 3, 4$

Impulse response: $h = \{1, -1, 2\}$

$$y[0] = x[-2]h[2] + x[-1]h[1] + x[0]h[0] = 2(1) = 2$$

$$y[1] = x[-1]h[2] + x[0]h[1] + x[1]h[0] = 2(-1) + 1(1) = -1$$

$$y[2] = x[0]h[2] + x[1]h[1] + x[2]h[0] = 2(2) + 1(-1) + 2(1) = 5$$

$$y[3] = x[1]h[2] + x[2]h[1] + x[3]h[0] = 1(2) + 2(-1) + 4(1) = 4$$

$$y[4] = x[2]h[2] + x[3]h[1] + x[4]h[0] = 2(2) + 4(-1) + 3(1) = 3$$

$$y[5] = x[3]h[2] + x[4]h[1] + x[2]h[0] = 4(2) + 3(-1) = 8 - 3 = 5$$

$$y[6] = x[4]h[2] + x[4]h[1] + x[2]h[0] = 3(2) = 6$$

So, the output $y[n] = \{2, -1, 5, 4, 3, 5, 6\}$

CONVOLUTION

Convolution is the operation that describes how systems act on signals.

Continuous-time Convolution:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

Discrete-time Convolution:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k] = x[n] * h[n]$$

CONVOLUTION

Once we know $h(t)$, we can find the output of the system for **any input** using **convolution**:

$$y(t) = x(t) * h(t)$$

The impulse response contains all the information about how the system behaves.

Convolving the input signal $x[n]$ with the system's impulse response $h[n]$ is referred to as **filtering** the input – a fundamental process in **speech production modelling**, extensively used in almost every speech processing application.

Filtering(Convolution with the system's impulse response):

1. Decompose the input signal into tiny impulses.
2. Replace each impulse with the corresponding scaled and shifted impulse response.
3. Sum all these responses to obtain the output signal.

CONVOLUTION PROPERTIES

Commutative: [whether we treat $x[n]$ as input or $h[n]$ as system, the output remains the same.]

$$x[n] * h[n] = h[n] * x[n]$$

Associative: $x[n] * (h[n] * g[n]) = (x[n] * h[n]) * g[n]$

Distributive: [Convolution distributes over addition.]

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

Shift Property: If $x[n - n_0]$ is a shifted version of $x[n]$, then:

$$x[n - n_0] * h[n] = y[n - n_0] \text{ [time-invariance preserved]}$$

Impulse Property: [Convolution with a unit impulse leaves the signal unchanged.]

$$x[n] * \delta[n] = x[n]$$

Scaling Property: [Scaling the input scales the output by the same factor.]

$$(a \cdot x[n]) * h[n] = a \cdot (x[n] * h[n])$$

Finite Duration Property: If $x[n]$ has length N_1 and $h[n]$ has length N_2 , then $y[n]$ has length $N_1 + N_2 - 1$.

CONVOLUTION

Exercises:

1. $x[n] = \{1, 2, 1, 3\}$, $h[n] = \{1, 0, -1\}$.

Determine output length and compute all $y[n]$

2. $x[n] = \{1, 2, 3\}$, $h[n] = \{1, -1, 2\}$.

Compute $y[n]$ and $y'[n]$ for $x[n-1]$; verify $y'[n] = y[n-1]$.

3. $x[n] = \{2, -1, 3\}$, $h[n] = \{1, -2, 1\}$.

Compute $y[n]$ and verify commutative property.

4. Express $x[n] = \{3, 4, 5\}$ as a sum of impulses and use convolution with $h[n] = \{1, 2\}$ to find $y[n]$.

5. $x[n] = \{1, 2\}$, $h_1[n] = \{1, 1\}$, $h_2[n] = \{1, -1\}$.

Show $x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$.

LIMITATIONS OF TIME-DOMAIN

Complex convolution: Convolution in time domain is computationally heavy. Requires summing many shifted and scaled versions of signals.

Limited insight into filtering: Difficult to visualize filter response. Frequency-domain plots (magnitude and phase) are more intuitive than time-domain.

No direct spectral interpretation: Cannot easily measure bandwidth, resonance, or spectral energy. Need Fourier or Z-transform for spectral insight.

Harder system characterization: Time-domain impulse response doesn't reveal poles, zeros, or stability clearly. Frequency-domain analysis simplifies system design.

Poor frequency visibility: In time domain, complex signals look irregular. Hard to identify frequency components (e.g., harmonics, formants). Speech or music signals appear cluttered in time domain.